

Pushty Shrimankar

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Research Profile

I am a theoretical physicist driven by the goal of uncovering the physics of the early universe through gravitational waves. My research bridges the gap between complex inflationary modeling, specifically magnetogenesis, and observational cosmology. As the lead author of arXiv:2604.04436, I combine analytical theory with computational tools like MCMC and GPR to probe cosmic anisotropies. I am eager to apply my background in sourced stochastic backgrounds to the next generation of GW observatories.

Education

M.Sc. Physics, Vellore Institute of Technology, Vellore 2024–2026

Current CGPA: 9.01

Master's Thesis: Indian Institute of Astrophysics (IIA)

Relevant Coursework: General Relativity, Astrophysics, Statistical Mechanics, Electrodynamics, Classical Mechanics, Mathematical Physics

B.Sc. Physics, H. and H. B. Kotak Institute of Science 2019–2022

9th Rank, Saurashtra University

Final-Year Project: Solar Differential Rotation (Prof. Hari Om Vats) — analyzed solar photospheric rotation using sunspot tracking.

Research Interests

Theoretical Astrophysics; Gravitational Waves; Fast Radio Bursts & CMB; Modified Gravity; Early-Universe Cosmology, Plasma Astrophysics.

Research Experience

Research Intern (Master's Thesis), Indian Institute of Astrophysics (IIA), Bengaluru 2025–Present

Supervisor: **Dr. Debika Chowdhury**

- **Core Work:** Modeled the Stochastic Gravitational-Wave Background (SGWB) sourced by primordial magnetic fields in non-helical configurations.
- **Key Result:** Successfully computed the spectral energy density $\Omega_{GW}(f)$, identifying parameter spaces where sourced signals are detectable by future observatories including **LISA, ET, and DECIGO**.
- **Technical Achievement:** Developed analytical and numerical solutions for magnetogenesis beyond the slow-roll regime, incorporating rigorous backreaction analysis to ensure physical consistency.

Cosmic Distance Duality Relation (CDDR) and BAO Analysis, VIT 2025–2026

- **Publication:** Lead author of [arXiv:2604.04436](https://arxiv.org/abs/2604.04436); engineered model-independent tests of the Cosmic Distance Duality Relation (CDDR) to probe directional anisotropies.

- Developed a Python-based analysis pipeline using NumPy, Astropy, and scikit-learn to reconstruct the duality ratio $\eta(z)$ from **Pantheon+** and **SH0ES** datasets.
- Employed **Gaussian Process Regression (GPR)** and **MCMC sampling (emcee)** for rigorous uncertainty propagation and dipole statistical inference to assess large-scale asymmetries.
- Cross-validated results with **DESI DR2 BAO** angular diameter distances to provide joint expansion constraints and test departures from the standard Λ CDM framework.
- Demonstrated consistency with the Etherington reciprocity theorem within observational limits, identifying potential signals of cosmic anisotropies in SN Ia positional distributions.

Similarity Parameters in Plasma and Gravitational Waves, VIT 2025

- Explored analogies between plasma wave propagation and gravitational wave signatures via self-similar scaling laws.

ICTS Summer School in Gravitational Waves 2025

- Lectures and hands-on sessions on GW detection pipelines, waveform modeling and mock-signal analysis using Python.

Gravitational Lensing, IUCAA Summer School 2021

- Introductory modules on lensing, GW fundamentals, and cosmological time-dilation concepts.

Stellar and Solar Observations, Neogen Solar and Stellar Observatory, Rajkot 2024–Present

- Operated 8–12 inch reflector telescopes for Messier object tracking and solar spectrographic observations.

Programs & Workshops

- ICTS Summer School in Gravitational Waves (2025)
- IUCAA Summer School in Astronomy and Astrophysics (2021)
- Photonics Lecture Series, IIT Guwahati (2021, online)

Technical Skills

- **Programming:** Python, Mathematica
- **Libraries/Tools:** NumPy, scikit-learn, Astropy, Matplotlib
- **Cosmology & GW Tools:** Cobaya sampler, HEALPix / Mollweide projection, LALSuite, Random walkers for MCMC inference
- **Methods:** Gaussian Process Regression (GPR), MCMC (**emcee**), Bayesian inference, Perturbation Theory

Achievements & Activities

- Ranked 9th in B.Sc. Physics, Saurashtra University
- Member, Big Bang Astronomy Club, Rajkot and Annual participant in Messier Marathon observations
- **Science Outreach:** Volunteered for the **Indian Institute of Astrophysics (IIA) Open Day**, communicating fundamental astrophysics concepts to the general public and students.